

PSY 4802- Pretest Fall 2026

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PSY 4802 Pretest

The goal of this assignment is to help me understand our collective R programming knowledge. It is graded on attempt, not accuracy. Because of this, please answer honestly and skip any questions you are unsure of how to answer. If you have an idea of how to solve the problem, please write your thought process. I'll use everyone's answers to determine pacing of the course.

Questions

1. Evaluate the following expressions:

a) 4 minus 3 times 2

```
(4-3)*2
```

```
## [1] 2
```

b) the square root of 125 divided by five squared

```
sqrt(125)/5^2
```

```
## [1] 0.4472136
```

c) find the sum of 7 and 6 and multiply by 13

```
(7+6)*13
```

```
## [1] 169
```

2) create variables that contain the following information:

a) a variable containing your favorite number

```
fav_num <- 8  
fav_num
```

```
## [1] 8
```

b) a list containing your top three favorite foods

```
fav_food <- list("Burger", "Pizza", "Milkshake")  
fav_food
```

```
## [[1]]  
## [1] "Burger"  
##  
## [[2]]  
## [1] "Pizza"  
##
```

```
## [[3]]
## [1] "Milkshake"
```

c) a vector with the numbers between 17 and 350

```
vector_1 <- c(17:350)
vector_1
```

```
##      [1]  17  18  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34
##     [19]  35  36  37  38  39  40  41  42  43  44  45  46  47  48  49  50  51  52
##     [37]  53  54  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70
##     [55]  71  72  73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88
##     [73]  89  90  91  92  93  94  95  96  97  98  99 100 101 102 103 104 105 106
##     [91] 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124
##    [109] 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142
##    [127] 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160
##    [145] 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178
##    [163] 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196
##    [181] 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214
##    [199] 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232
##    [217] 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250
##    [235] 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268
##    [253] 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286
##    [271] 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304
##    [289] 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322
##    [307] 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340
##    [325] 341 342 343 344 345 346 347 348 349 350
```

d) a vector with odd numbers between 67 and 137

```
# Don't know
```

e) a list with the words “I can do this” and “I can’t do this” alternating six times

```
# Don't know
```

3) Evaluate the following logical statements

a) 3 times 4 is not equal to 6 plus 7

```
3*4 != 6+7
```

```
## [1] TRUE
```

b) 4 squared is greater than 3 and 5 is less than 9

```
4^2 > 3 & 5 < 9
```

```
## [1] TRUE
```

c) 12 divided by 2 is less than 5 or 4 is greater than 3

```
12 / 2 < 5 | 4 > 3
```

```
## [1] TRUE
```

4) import the world happiness survey 2025 data (see assignment in R)

```
library(readr)
worldhappinessreport2025 <- read_csv("worldhappinessreport2025.csv")
```

```
## Rows: 147 Columns: 9
```

```
## -- Column specification -----
```

```
## Delimiter: ","
## chr (1): countryName
## dbl (8): ladderScore, GDP, socialSupport, healthyLifeExpectancy, lifeChoices...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

5) print the column names in the dataset

```
colnames(worldhappinessreport2025)
```

```
## [1] "countryName"      "ladderScore"
## [3] "GDP"              "socialSupport"
## [5] "healthyLifeExpectancy" "lifeChoices"
## [7] "generosity"       "perceptionsofCorruption"
## [9] "dystopia"
```

6) summarize the generosity column

```
summary(worldhappinessreport2025$generosity)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0665  0.1110  0.1134  0.1505  0.3230
```

7) find the average perceived corruption value

```
#mean(worldhappinessreport2025$perceptionsofCorruption)
summary(worldhappinessreport2025$perceptionsofCorruption)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
## 0.0000  0.0665  0.1180  0.1468  0.1845  0.5220      1
```

8) Open the mtcars dataset

```
data("mtcars")
head(mtcars)
```

```
##           mpg  cyl  disp  hp  drat    wt  qsec vs  am  gear  carb
## Mazda RX4      21.0   6  160 110 3.90 2.620 16.46 0   1    4    4
## Mazda RX4 Wag  21.0   6  160 110 3.90 2.875 17.02 0   1    4    4
## Datsun 710     22.8   4  108  93 3.85 2.320 18.61 1   1    4    1
## Hornet 4 Drive  21.4   6  258 110 3.08 3.215 19.44 1   0    3    1
## Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02 0   0    3    2
## Valiant        18.1   6  225 105 2.76 3.460 20.22 1   0    3    1
```

9) describe the data in the mtcars dataset

32 cars compared by things such as MPG, engine displacement, horsepower, and weight.

10) What kind of variable is present in the am column? How do you know?

Probably a boolean because it is just 1 and 0 so it is likely a True/False category.

11) how many values are in the dataset?

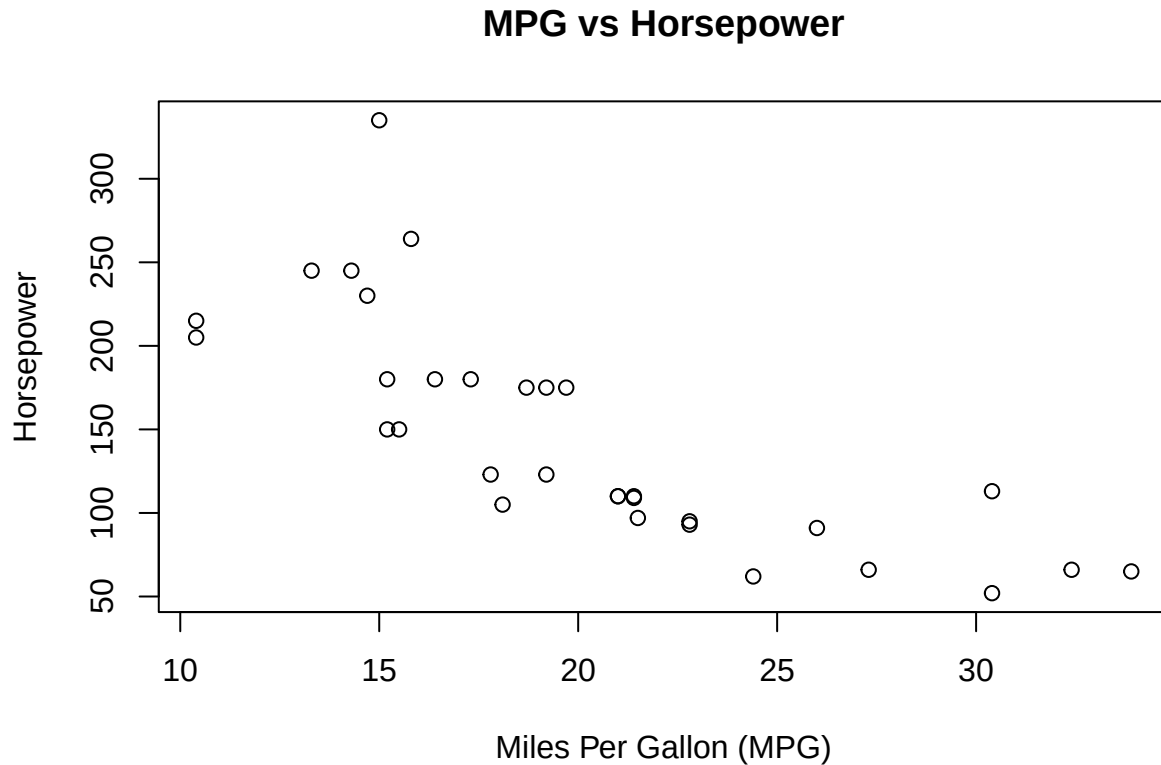
32 cars and 11 variables so 352

12) create a scatterplot of mpg vs hp

I used ggplot2 once, but I don't remember how. I tried using the suggested shortcuts but that didn't really work. I tried looking at the documentation, and then I realized I didn't want to didn't want to. I just used plot().

```
#ggplot2::aes(mtcars$mpg,mtcars$hp)
```

```
plot(mtcars$mpg, mtcars$hp,
     main = "MPG vs Horsepower",
     xlab = "Miles Per Gallon (MPG)",
     ylab = "Horsepower")
```



13) Add a regression line to your graph in 12

#Don't know

14) create a bar graph of the gear column

#Don't know

15) open the chickweight dataset

```
data("ChickWeight")
head(ChickWeight)
```

```
##   weight Time Chick Diet
## 1    42    0     1    1
## 2    51    2     1    1
## 3    59    4     1    1
## 4    64    6     1    1
## 5    76    8     1    1
## 6    93   10     1    1
```

16) What format is the chickweight dataset in?

I do not know

17) create a subset of data for Chick 4

#Don't know

18) change the chickweight dataset from long to wide

#Don't know